

Somanshu Mohapatra

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SKILLS

Languages:	C++, Python, Java, JavaScript, HTML, SQL, Embedded C, MATLAB, GNU Octave.
AI & Vision:	AIML, Deep Learning (PyTorch, TensorFlow, CNN, YOLO), OpenCV, OpenVINO, AI Image Generation (Stable Diffusion, AUTOMATIC1111).
Robotics:	Forward/Inverse Kinematics (DH Parameters), ROS2, RViz2, MoveIt, Gazebo, Path Planning, Motion Control, PID Control, Sensor Fusion, SLAM.
Electronics:	Arduino UNO, Raspberry Pi, STM32, ATmega328P, RP2040, Motor Drivers, Serial Communication (UART, I2C, SPI), LabVIEW, Proteus, NGSpice.
CAD & Design:	SolidWorks, Fusion 360, AutoCAD, Creo Parametric, ANSYS HFSS, Rapid Prototyping, 3D Printing (FDM/SLA), Bambu Slicer, LightBurn.
Soft Skills:	Leadership, Strategic Planning, Cross-functional Collaboration, Analytical Problem-Solving.

TRAININGS

AutomateX: Mastering PLCs for Industrial Automation LPU	Mar' 25
- Gained expertise in PLC architecture, ladder logic, and industrial automation standards. - Modeled automated workflows in TwidoSuite, managing sensor integration and high-power actuators. - Validated fail-safe protocols and sequence patterns to ensure operational reliability.	

PROJECTS

2.4GHz Microstrip Patch Antenna ANSYS HFSS	Oct' 25
- Designed a high-frequency antenna for 2.4GHz wireless communication standards. - Performed electromagnetic field analysis via ANSYS HFSS to assess return loss, gain, and radiation behavior. - Refined substrate selection and dimensions to maximize impedance matching and antenna efficiency.	
Dynamic Audio Monitoring and Narration Raspberry Pi, Python, AssemblyAI API	May' 25
- Engineered a real-time transcription system for the hearing impaired using Raspberry Pi. - Integrated AssemblyAI API and WebSockets with an I2C SSD1306 OLED display using multi-threading. - Reduced latency by deploying an Intel AI Accelerator on Raspberry Pi 4B for accelerated inference.	
3-DOF Robotic Arm with Real-Time Control MATLAB, SolidWorks, Arduino	Apr' 24
- Fabricated a 3-DOF arm using 3D-printed components to improve the payload-to-weight ratio. - Configured real-time motor control via MATLAB-UART communication for precise joint actuation. - Developed a framework for future Forward and Inverse Kinematics integration to enable automated trajectory planning.	
Python-SQL Accounting Software Python, MySQL, CLI	Jun' 23
- Built a database-driven CLI application for financial record and transaction management. - Executed CRUD operations using Python-MySQL connectivity to ensure efficient data handling. - Implemented search functionality by client name and date, streamlining data retrieval.	

CERTIFICATES

Data Base Management System NPTEL	Aug' 25
Fusioncraft Resolute Lab	Mar' 24

ACHIEVEMENTS

Patent: Method for Enhancing Food Accessibility and Energy Efficiency in Refrigeration	Sep' 25
Application No.: 202511085342 A	

EDUCATION

Lovely Professional University	Phagwara, Punjab
BTech Robotics and Automation; CGPA: 8.38	Aug' 23 – Present
Shri S.N. Siddeshwar Public School	Gurugram, Haryana
Intermediate, PCMB; Percentage: 85.6%	Mar' 21 – May' 23
Ryan International School	Gurugram, Haryana
Matriculation; Percentage: 81.6%	Mar' 20 – May' 21